Task: Network VAPT Report
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Executive Summary

Network Vulnerability Assessment and Penetration Testing (VAPT) is the process of identifying, analyzing, and exploiting security weaknesses within an organization's network infrastructure, including servers, devices, services, and communication channels. It involves both automated scanning and manual testing to detect vulnerabilities such as open ports, misconfigurations, weak authentication mechanisms, outdated services, and unpatched systems.

The assessment is typically performed following industry standards like the **OWASP** and **NIST**, using tools such as **Nmap**, **Metasploit**, and **Wireshark**.

CVSS Scoring System:

The Common Vulnerability Scoring System (CVSS) provides a standardized method for assessing vulnerability severity. Security analysts often use the CVSS calculator provided by the National Vulnerability Database to determine vulnerability scores.

Score Range	Risk Level
0 – 3.9	**Low**
4 – 6.9	**Medium**
7 – 8.9	**High**
9 – 10	**Critical**

Risk Matrix:

A risk matrix is another method used to prioritize vulnerabilities based on their likelihood and impact. EX:,

Vulnerability	Likelihood	Impact	Risk Level
SQL Injection	High	High	Critical
XSS	Medium	Medium	Medium
Open Ports	Low	Medium	Low

Information:

Item	Details
Project Title	Network Vulnerability Assessment of Metasploitable and Kali Linux Environment
Assessment Type	Vulnerability Assessment Lab
Tools Used	Kali Linux, Nmap, Ping, Ettercap
Testing Environment	VirtualBox Lab
Date	03/04/2026

Executive summary:

The objective of this assessment was to identify security vulnerabilities in a controlled testing environment using open-source security tools. The assessment was performed on a vulnerable virtual machine to simulate a real-world security evaluation scenario.

The scanning process identified several vulnerabilities including outdated software versions, exposed services, and configuration weaknesses. These issues could potentially allow attackers to gain unauthorized access to the system if not properly mitigated. Risk assessment was performed using CVSS scoring to prioritize remediation efforts.

The below table gives a brief description of each risk rating visualized in this report.

Impact	CVSS Score	Definition
Critical	9.0 – 10.0	Critical vulnerabilities are the highest level of security risks, demanding immediate attention and mitigation within an organization. They pose an imminent and severe threat, with the potential for unauthorized access, data breaches, or significant system compromise. Swift

		remediation is crucial to maintain robust security.
High	7.0 – 8.9	High severity vulnerabilities should be considered a top priority in terms of mitigation. These are the most severe issues and generally cause an immediate security concern to the enterprise
Medium	4.0 – 6.9	Medium severity vulnerabilities are a lower priority but should still be remediated in a timely manner. These are moderate exploits that have less of an impact on the environment.
Low	0.1 – 3.9	Low severity vulnerabilities are real but trivially impactful to the environment. These should only be remediated after the HIGH and MEDIUM vulnerabilities are resolved.
Informational	0.0	Informational vulnerabilities have no impact as such to the environment by themselves. However, they might provide an attacker with information to exploit other vulnerabilities.

The following control areas are the easiest to exploit or expose the environment to the highest level of risk:

Sr. No.	Vulnerability Name	Severity Level
1	Cleartext Credential Transmission over HTTP	**Critical**

Vulnerabilities Deatil:

Sr.No. 1	**Name of Vulnerability:** Cleartext Credential Transmission over HTTP	**CWE Details:** CWE-319
Critical	**CVSS Score:** 9.1	
Vulnerable URL: DVWA, PhpMyAdmin		
Description: The application transmitting sensitive authentication data (usernames and passwords) over **unencrypted HTTP** instead of HTTPS.		
Impact: • **Credential theft** (usernames & passwords exposed) • **Session hijacking** via stolen cookies • **Unauthorized access** to databases or admin panels • Potential **privilege escalation** (e.g., root login observed) • Susceptibility to Man-in-the-Middle (MitM) attacks		
PoC:		

http.request.method

No.	Time	Source	Destination	Protocol	Length	Info
4	0.022268374	192.168.1.100	192.168.1.200	HTTP	442	GET /phpMyAdmin/ HTTP/1.1
16	2.847074786	192.168.1.100	192.168.1.200	HTTP	436	GET /dvwa/ HTTP/1.1
20	3.290191138	192.168.1.100	192.168.1.200	HTTP	661	GET /phpMyAdmin/phpmyadmin.css.php?lang=en-utf-8&convcharset=utf-8&token=9
25	3.346409439	192.168.1.100	192.168.1.200	HTTP	512	GET /dvwa/login.php HTTP/1.1
79	4.126481603	192.168.1.100	192.168.1.200	HTTP	626	GET /phpMyAdmin/themes/original/img/logo_right.png HTTP/1.1
90	4.987233567	192.168.1.100	192.168.1.200	HTTP	567	GET /phpMyAdmin/print.css HTTP/1.1
94	5.081733498	192.168.1.100	192.168.1.200	HTTP	457	GET /dvwa/dvwa/css/login.css HTTP/1.1
97	5.138838771	192.168.1.100	192.168.1.200	HTTP	519	GET /dvwa/dvwa/images/login_logo.png HTTP/1.1
98	5.189958449	192.168.1.100	192.168.1.200	HTTP	765	GET /phpMyAdmin/themes/original/img/s_warn.png HTTP/1.1
101	5.202569326	192.168.1.100	192.168.1.200	HTTP	767	GET /phpMyAdmin/themes/original/img/s_notice.png HTTP/1.1
123	5.325824871	192.168.1.100	192.168.1.200	HTTP	620	GET /phpMyAdmin/favicon.ico HTTP/1.1
141	5.748476405	192.168.1.100	192.168.1.200	HTTP	489	GET /twiki/ HTTP/1.1
160	22.353238087	192.168.1.100	192.168.1.200	HTTP	494	GET /mutillidae/ HTTP/1.1
161	22.353673293	192.168.1.100	192.168.1.200	HTTP	487	GET /dav/ HTTP/1.1
166	23.176442366	192.168.1.100	192.168.1.200	HTTP	1017	POST /phpMyAdmin/index.php HTTP/1.1 (application/x-www-form-urlencoded)
173	23.399602197	192.168.1.100	192.168.1.200	HTTP	478	GET /icons/blank.gif HTTP/1.1
174	23.400741367	192.168.1.100	192.168.1.200	HTTP	477	GET /icons/back.gif HTTP/1.1
219	24.833700139	192.168.1.100	192.168.1.200	HTTP	451	GET /mutillidae/styles/global-styles.css HTTP/1.1
220	24.843887605	192.168.1.100	192.168.1.200	HTTP	463	GET /mutillidae/styles/ddsmoothmenu/ddsmoothmenu.css HTTP/1.1
227	24.886785316	192.168.1.100	192.168.1.200	HTTP	465	GET /mutillidae/styles/ddsmoothmenu/ddsmoothmenu-v.css HTTP/1.1
230	24.893368144	192.168.1.100	192.168.1.200	HTTP	439	GET /mutillidae/javascript/bookmark-site.js HTTP/1.1
245	25.035946975	192.168.1.100	192.168.1.200	HTTP	451	GET /mutillidae/javascript/ddsmoothmenu/ddsmoothmenu.js HTTP/1.1
260	25.092647033	192.168.1.100	192.168.1.200	HTTP	449	GET /mutillidae/javascript/ddsmoothmenu/jquery.min.js HTTP/1.1
318	25.340653201	192.168.1.100	192.168.1.200	HTTP	510	GET /mutillidae/images/owasp-logo-400-300.png HTTP/1.1
319	25.341873375	192.168.1.100	192.168.1.200	HTTP	505	GET /mutillidae/images/coykillericon.png HTTP/1.1

```
▸ Frame 20: Packet, 661 bytes on wire (5288 bits), 661 bytes captured (5288 bits) on interface eth0, id 0
▸ Ethernet II, Src: PCSSystemtec_d1:f8:5d (08:00:27:d1:f8:5d), Dst: PCSSystemtec_fa:d6:3c (08:00:27:fa:d6:3c)
▸ Internet Protocol Version 4, Src: 192.168.1.100, Dst: 192.168.1.200
▸ Transmission Control Protocol, Src Port: 34980, Dst Port: 80, Seq: 747, Ack: 5544, Len: 595
▾ Hypertext Transfer Protocol
  ▸ GET /phpMyAdmin/phpmyadmin.css.php?lang=en-utf-8&convcharset=utf-8&token=95947ee7a75916166471124a82f24b19&js_frame=right&nocach
    Host: 192.168.1.200\r\n
    User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0\r\n
    Accept: text/css,*/*;q=0.1\r\n
    Accept-Language: en-US,en;q=0.5\r\n
    Accept-Encoding: gzip, deflate\r\n
    Connection: keep-alive\r\n
    Referer: http://192.168.1.200/phpMyAdmin/\r\n
  ▾ Cookie: phpMyAdmin=363851cd9fd5b2a0aece1260076ae529b04ac893; pma_lang=en-utf-8; pma_charset=utf-8; pma_theme=original; PHPSESSI
      Cookie pair: phpMyAdmin=363851cd9fd5b2a0aece1260076ae529b04ac893
      Cookie pair: pma_lang=en-utf-8
      Cookie pair: pma_charset=utf-8
      Cookie pair: pma_theme=original
      Cookie pair: PHPSESSID=12dfbe320da89bafedb10e51d311c84f
    Priority: u=2\r\n
    \r\n
    [Response in frame: 77]
    [Full request URI: http://192.168.1.200/phpMyAdmin/phpmyadmin.css.php?lang=en-utf-8&convcharset=utf-8&token=95947ee7a759161664
```

```
0040  84 85 47 45 54
0050  6e 2f 70 68 70
0060  2e 70 68 70 3f
0070  2d 38 26 63 6f
0080  74 66 2d 38 26
0090  65 65 37 61 37
00a0  32 34 61 38 32
00b0  72 61 6d 65 3d
00c0  68 65 3d 32 34
00d0  54 50 2f 31 2e
00e0  32 2e 31 36 38
00f0  72 2d 41 67 65
0100  2f 35 2e 30 20
0110  20 78 38 36 5f
0120  30 29 20 47 65
0130  31 20 46 69 72
0140  0a 41 63 63 65
0150  73 2c 2a 2f 2a
0160  65 70 74 2d 4c
0170  2d 55 53 2c 65
0180  63 65 70 74 2d
0190  7a 69 70 2c 20
01a0  6e 6e 65 63 74
01b0  6c 69 76 65 0d
```

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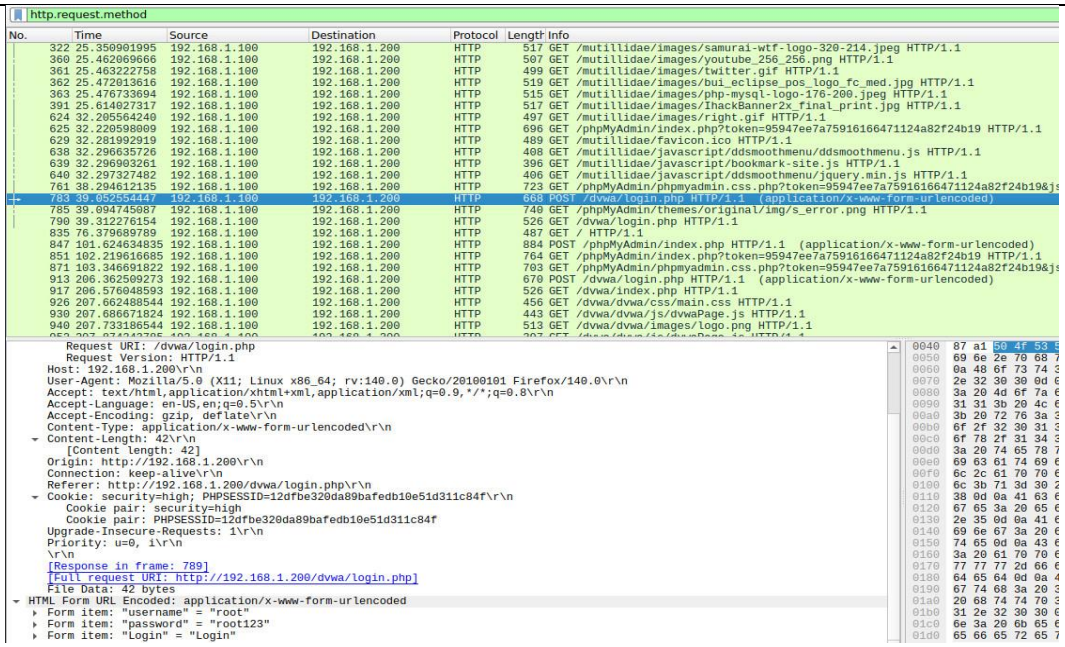
220	24.843887605	192.168.1.100	192.168.1.200	HTTP	463	GET /mutillidae/styles/ddsmoothmenu/ddsmoothmenu.css HTTP/1.1
227	24.886785316	192.168.1.100	192.168.1.200	HTTP	465	GET /mutillidae/styles/ddsmoothmenu/ddsmoothmenu-v.css HTTP/1.1
230	24.893368144	192.168.1.100	192.168.1.200	HTTP	439	GET /mutillidae/javascript/bookmark-site.js HTTP/1.1
245	25.035946975	192.168.1.100	192.168.1.200	HTTP	451	GET /mutillidae/javascript/ddsmoothmenu/ddsmoothmenu.js HTTP/1.1
260	25.092647033	192.168.1.100	192.168.1.200	HTTP	449	GET /mutillidae/javascript/ddsmoothmenu/jquery.min.js HTTP/1.1
318	25.340653201	192.168.1.100	192.168.1.200	HTTP	510	GET /mutillidae/images/owasp-logo-400-300.png HTTP/1.1
319	25.341873375	192.168.1.100	192.168.1.200	HTTP	505	GET /mutillidae/images/coykillericon.png HTTP/1.1

```
    Connection: keep-alive\r\n
    Referer: http://192.168.1.200/phpMyAdmin/\r\n
  ▾ Cookie: phpMyAdmin=363851cd9fd5b2a0aece1260076ae529b04ac893; pma_lang=en-utf-8; pma_charset=utf-8; pma_theme=original; pma_f
      Cookie pair: phpMyAdmin=363851cd9fd5b2a0aece1260076ae529b04ac893
      Cookie pair: pma_lang=en-utf-8
      Cookie pair: pma_charset=utf-8
      Cookie pair: pma_theme=original
      Cookie pair: pma_fontsize=82%25
      Cookie pair: PHPSESSID=12dfbe320da89bafedb10e51d311c84f
    Upgrade-Insecure-Requests: 1\r\n
    Priority: u=0, i\r\n
    \r\n
    [Response in frame: 565]
    [Full request URI: http://192.168.1.200/phpMyAdmin/index.php]
    File Data: 271 bytes
▾ HTML Form URL Encoded: application/x-www-form-urlencoded
  ▸ Form item: "phpMyAdmin" = "363851cd9fd5b2a0aece1260076ae529b04ac893"
  ▸ Form item: "phpMyAdmin" = "363851cd9fd5b2a0aece1260076ae529b04ac893"
  ▸ Form item: "pma_username" = "test"
  ▸ Form item: "pma_password" = "test"
  ▸ Form item: "server" = "1"
  ▸ Form item: "phpMyAdmin" = "363851cd9fd5b2a0aece1260076ae529b04ac893"
  ▸ Form item: "lang" = "en-utf-8"
  ▸ Form item: "convcharset" = "utf-8"
  ▸ Form item: "token" = "95947ee7a75916166471124a82f24b19"
```

```
0040  86 3e 50 4f 53
0050  69 6e 2f 69 6e
0060  50 2f 31 2e 31
0070  2e 31 36 38 2e
0080  2d 41 67 65 6e
0090  35 2e 30 20 28
00a0  78 38 36 5f 36
00b0  29 20 47 65 63
00c0  20 46 69 72 65
00d0  41 63 63 65 70
00e0  6c 2c 61 70 70
00f0  74 6d 6c 2b 78
0100  69 6f 6e 2f 78
0110  2a 3b 71 3d 30
0120  4c 61 6e 67 75
0130  65 6e 3b 71 3d
0140  2d 45 6e 63 6f
0150  29 64 65 66 6c
0160  74 2d 54 79 70
0170  69 6f 6e 2f 78
0180  72 6c 65 6e 63
0190  6e 74 2d 4c 65
01a0  4f 72 69 67 69
01b0  39 32 2e 31 36
01c0  6e 6e 65 63 74
01d0  6c 69 76 65 0d
```



Recommendation:

- Enforce HTTPS
- Secure Cookies
- Disable HTTP Access
- Implement HSTS
- Use Strong Authentication Controls
- Network Security

Reference: https://cwe.mitre.org/data/definitions/319.html